Activity

① CCTV image contains Salt and pepper noise:

Noise: Salt and pepper Noise.

Technique: Median filter because it remove impulse noise while preserving edges.

② Satellite image is corrupted by Gaussian noise:

Noise: Gaussian Noise

technique: Gaussian filter, because it smooths Gaussian noise effectively.

③ Medical X-ray image has random noise but fine edge must be preserved.

Noise: Gaussian Noise.

Technique: Bilateral filter, because it reduces noise while preserving edges.

④ Scanned document has black and white dots:

Noise: salt and pepper noise.

Technique: Median filter, because it removes black and white impulse dots.

⑤ Noise: Grain (speckle noise.

Technique: Non-local means filter, because it remove grain while preserving details.

(6) foggy road image has very low contrast :

Problem: low contrast (not noise). Haze No...

Technique : Histogram equalization - improves contrast and visibility.

(7) underwater image

Noise : Scattering Noise

Technique: Histogram equalization - enhances Contrast and image clarity.

(8) Chest X-ray

Noise: low contrast Noise

Technique: Histogram equalization - increases brightness and intensity variation.

(9) Night - time traffic image

Noise: low illumination Noise

Technique: Histogram Equalization - increases brightness and intensity variation.

(10) Grayscale image with narrow intensity range.

Noise: low contrast noise

Technique : Histogram Equilization → Spreads intensity value for better image quality.